

HYDROTHERMAL VENTS

Volcanic midocean ridges are peppered with submarine geysers that erupt hot water full of dissolved chemicals. These hydrothermal vents support dense communities of animals that grow much faster, and bigger, than most deep ocean life. This is because they do not depend on the little food that drifts down from the surface. They have their own food supply, created by using the chemical energy of the vents themselves. This ecosystem is one of the few on Earth that does not rely on sunlight.

◀ CHEMICAL CLOUDS

The hot water that pours from hydrothermal vents contains dissolved metal sulfides that turn into solid, sooty particles when they mix with cold ocean water. The particles can make the water look like black smoke, so the vents are often called black smokers. Some of the chemicals are lethal to most marine life, but the organisms that live around black smokers not only survive them, but also thrive on them.

hydrothermal
vents

MAKING FOOD



The rocks around hydrothermal vents are covered with dense white mats of microscopic bacteria. These are able to absorb the normally toxic hydrogen sulfide in the hot vent water and combine it with oxygen in a chemical reaction that produces energy. They use the energy to make sugar from water and dissolved carbon dioxide, in a process called chemosynthesis. This is like photosynthesis, but it does not need light. It is likely that some of the earliest forms of life on Earth made food in the same way.

POMPEII WORMS ▶

The water that gushes from hydrothermal vents can be astonishingly hot, reaching 850°F (450°C) or more. Yet some animals can live surprisingly close to the superheated water. The 4 in (10 cm) long Pompeii worm lives on the chimneys that build up around vents, with its head in water at a temperature of about 70°F (20°C), but its tail in water heated to 160°F (70°C) or more. This would kill any other animal.

